

# Well Model

## Injector

The injection well model correlates the reservoir flow rate of phase  $j$  ( $j = g, w$ ) to the wellbore pressure,  $p_{bh}$  and the pressure at grid point via the relationship:

$$Q_j = \sum_l W I_l \lambda_{T,l} (p_{bh} - p_{o,i}) \quad j = g, w \quad (A2.1)$$

$$W I = 2 \pi \text{ ff } k h \frac{wfrac}{\ln(r_e/r_w) + S} \quad j = g, w \quad (A2.2)$$

with  $p_{bh} > p_{o,i}$

where

$Q_j$  = flow rate of phase  $j$  ( $j = g, w$ ) at reservoir conditions (m<sup>3</sup>/day | ft<sup>3</sup>/day)

$p_{bh}$  = bottomhole pressure (kPa | psia)

$p_{o,i}$  = pressure of  $i$ -th grid block containing the well (kPa | psia)

$W I_{j,1}$  = well injectivity index for phase  $j$  ( $j = g, w$ ) to layer 1

$wfrac$  = well fraction, governed by areal geometry (see [Figure A.1](#))

$k$  = effective permeability in the plane perpendicular to the well direction

$h$  = gridblock thickness in well direction (m | ft)

$$\lambda_T = \sum_j \frac{k_{rj}}{\mu_j} j = o, g, w \text{ total mobility of the fluid in the well block. The relative permeabilities are calculated using the gridblock saturation}$$

$$r_w = \text{wellbore radius (m | ft)}$$

$$r_e = CC \sqrt{A_i / wfrac} \text{ effective radius (m | ft)}$$

$$A_i = \text{area of } i\text{-th gridblock perpendicular to well direction (m}^2 \text{ | ft}^2\text{)}$$

$$CC = \text{geometric factor (see [Figure A.1](#))}$$

$$S = \text{skin factor (dimensionless)}$$

$$ff = \text{fraction of completion of the well in the gridblock}$$

$$\sum_l = \text{summation over all layers}$$

The well model is a generalization of the well model proposed by [Peaceman \(1987\)](#) and [Peaceman \(1983\)](#) for square and non-square gridblocks. The mobility treatment follows the suggestion of [Chappelear and Williamson \(1981\)](#). In addition, the geometric factor allows the determination of the equivalent radius from both the geometry of the gridblock and the location of the well in the gridblock.

## Producer

The production well model is similar to the preceding case. The flow rate of phase  $j$  ( $j = o, g, w$ ) at reservoir conditions is given by:

$$Q_j = \sum_l PI_{j,l} \left( \frac{k_r}{\mu} \right)_{j,l} (p_{o,i} - p_{bh}) \quad j = o, g, w \quad (\text{A2.3})$$

$$PI_j = 2 ff k h \frac{wfrac}{\ln(r_e/r_w) + S} \quad (\text{A2.4})$$

with  $p_{bh} < p_{o,i}$

where

$k_{rj}$  = relative permeability of phase  $j$  in the  $i$ -th gridblock (fraction), it is calculated using the gridblock saturations

$\mu_j$  = viscosity of phase  $j$  in the  $i$ -th gridblock, cp

$PI_j$  = well productivity index for phase  $j$  ( $j = o, g, w$ ) from layer 1

All the other parameters have the same definitions as in Equation [\(A2.2\)](#).

In many field simulation problems, the well is completed through several gridblocks and either the total rate or the bottomhole pressure of that multilayer well is specified. In these cases the summation in I must be carried out over all completion blocks. [Figure A.2](#) shows a multiblock well completion where for convenience both the gridblocks and well elements are numbered from 1 to  $n_c$ . The pressure in the  $i$ -th well element,  $p_{bh,i}$ , is related to the bottomhole pressure,  $p_{bh,1}$  by the following equation:

$$p_{bh,i} = p_{bh,1} - \int_{h_i}^{h_1} \tilde{\rho}^T g dh \quad (A2.5)$$

where  $\tilde{\rho}^T$  is the mass density of all phases ( $\text{kg/m}^3$ ) within the wellbore,  $g$  the gravitational acceleration ( $\text{m/s}^2$ ) and  $h$  the depth (m), measured positive downward.

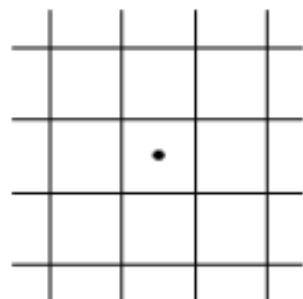
For 1-D slim tube runs, the following are used:

Injector

$$Q_j = CC k_x (\Delta y \Delta z) \lambda_T (p_{bh} - p_{o,i}) \text{ ff}$$

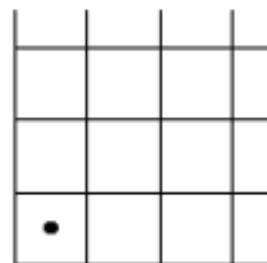
Producer

$$Q_j = CC\,k_y\,(\Delta y\Delta z)\,\left(\frac{k_r}{\mu}\right)_j\,(p_{o,i}-p_{bh})\,\text{ff}$$



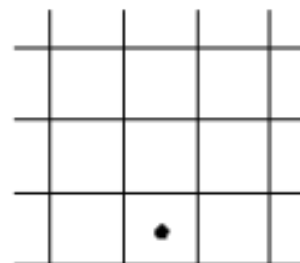
$f = 1$   
CC = 0.37

(a) Centre of Square Grid Block



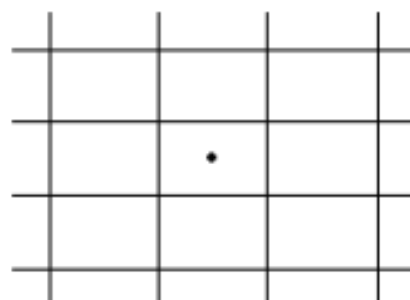
$f = 1$   
CC = 0.34

(b) Centre of Corner Block



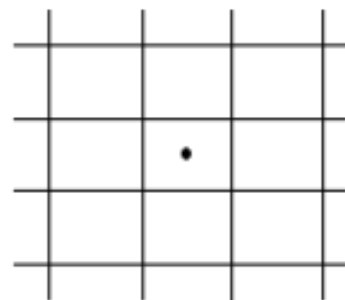
$f = 1$   
CC = 0.35

(c) Centre of Edge Block



$f = 1$  CC = 0.41

(d) Centre Rectangular Grid Block Ratio  $\Delta x/\Delta y=2$



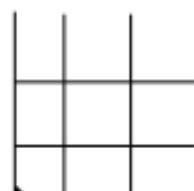
$f = 1$  CC = 0.39

(e) Centre Rectangular Grid Block Ratio  $\Delta x/\Delta y=3/2$



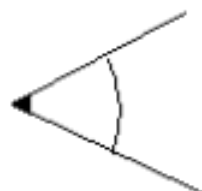
$f = 1/4$   
CC = 0.64

(f) Well in Corner



$f = 1/4$   
CC = 0.56

(g) Edge Blocks Half Size

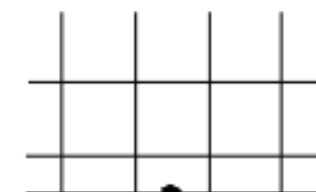


$f = \text{segment}$   
CC = 0.5

(h) Radial Segment  
 $2\pi$  radian  $f = 1$   
 $\pi$  radian  $f = 1$   
 $1/2\pi$  radian  $f = 1/4$



$f = 1/2$



$f = 1/2$

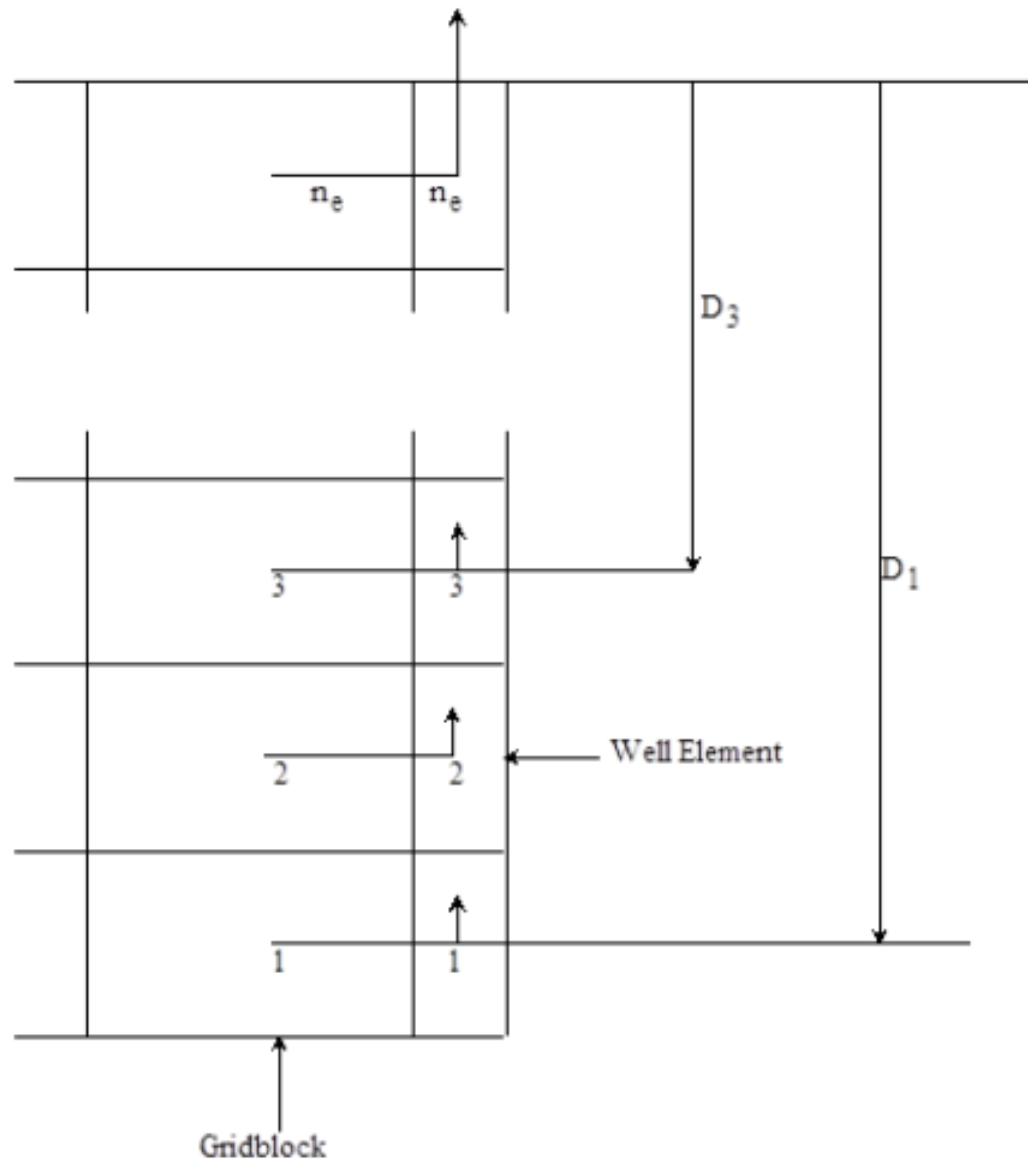
$\gamma = 1/2$   
CC = 0.54

$\gamma = 1/2$   
CC = 0.47

(i) Well on Edge

(j) Edge Blocks  
Half Size

Figure A.1: Well fraction and geometrical factor for various common geometries



*Figure A.2: Multiblock well completion*